THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 5: **Describing Motion Around Us** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A body covers 5 m in the first second and 15 m in the next second. This indicates: 1

(a) rest **(b)** uniform velocity
(c) retardation **(d)** uniform acceleration

2. A car travels at 90 km/h. Its speed in m/s is: 1

(a) 324 m/s **(b)** 9 m/s **(c)** 25 m/s **(d)** 90 m/s

3. The total distances covered in 1 s, 2 s, 3 s by a body falling from rest are in the ratio: 1

(a) 1 : 2 : 3 **(b)** 1 : 4 : 9 **(c)** 1 : 3 : 5 **(d)** 1 : 8 : 27

4. A cyclist stops pedalling and rolls to rest in 10 s from 5 m/s. The average retardation is: 1

(a) 0.5 m/s squared **(b)** 2 m/s squared **(c)** 50 m/s squared **(d)** 5 m/s squared

5. A body travels 40 m in 4 s and then 60 m in 6 s. Its average speed is: 1

(a) 5 m/s **(b)** 10 m/s **(c)** 20 m/s **(d)** 100 m/s

6. A ball is thrown upward at 30 m/s. The time to reach the highest point (g = 10) is: 1

(a) 3 s **(b)** 6 s **(c)** 30 s **(d)** 1.5 s

7. A jet starts from rest and reaches 100 m/s in 20 s. Its acceleration is: 1

(a) 80 m/s squared **(b)** 0.2 m/s squared **(c)** 2000 m/s squared **(d)** 5 m/s squared

8. A particle moves 3 m east and 3 m west in 6 s. Its average velocity is: 1

(a) 6 m/s **(b)** zero **(c)** 0.5 m/s **(d)** 1 m/s

9. For the above, the maximum height is: 1

(a) 45 m **(b)** 90 m **(c)** 15 m **(d)** 30 m

10. Which statement is correct about a body moving in a straight line with uniform acceleration? 1

(a) it is at rest
(b) its speed is constant
(c) its velocity changes by equal amounts in equal times
(d) its displacement is constant

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Draw in words the s-t graph of a body that moves 10 m in 10 s, rests for 10 s and returns in 10 s. 2

12. A body moves 6 m east and 8 m north. Find the displacement and the distance. 2

13. A car stops in 5 s from 15 m/s. Find the retardation and the stopping distance. 2

14. Two cars A and B start from the same point; A moves at 10 m/s and B at 15 m/s in the same direction. Find the distance between them after 10 s. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A train at 90 km/h brakes with retardation 1.25 m/s squared. Find the distance and time to stop. 3

16. A stone dropped from a bridge takes 2.5 s to hit the water. Find the height of the bridge and the impact speed (g = 10). 3

17. A body moving with uniform acceleration has velocity 5 m/s at t = 0 and 25 m/s at t = 10 s. Find the acceleration and the displacement. 3

OR

A person walks 40 m east, 30 m north and 40 m west in 50 s. Find the distance, displacement and average speed. 3

18. A body completes a circle of circumference 20 m in 5 s. Find the speed, the average velocity after half a revolution (displacement = diameter, taking pi = 3.14). 3

OR

Explain why a body moving on a circle at a constant speed has a changing velocity and acceleration directed to the centre. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A stone is dropped from the top of a tower of height 80 m and at the same time another stone is thrown up from the base at 40 m/s. When and where do they meet (g = 10)? 5

OR

A race: A runs 100 m in 12.5 s, B in 12 s. Find the speeds in m/s and km/h, and the distance by which B wins. 5

20. Derive the formula for the distance covered in the nth second and calculate it for a = 4 m/s squared, u = 2 m/s, n = 5. 5

OR

Explain the graphical derivation of the equation of motion s = ut + (1/2)at squared from the v-t graph and verify numerically with u = 4, a = 2, t = 5. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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